The documentation and process conversion measures necessary to comply with this revision shall be completed by 20 January 1994

INCH-POUND

MIL-S-19500/443A
20 October 1993
SUPERSEDING
MIL-S-19500/443
6 April 1971

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, SWITCHING
TYPE 1N5719 JANTX AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for silicon, pin diodes intended for use in switching applications in the frequency range of 10 MHz through 12 GHz. Two levels of product assurance are provided for each device type as specified in MIL-S-19500. For JAN quality level (see 6.3).

1.2 Physical dimensions. See figure 1.

1.3 Maximum ratings.

V_{RWM}	V_{RWM} (non-repetitive)	P_T 1/	T_{STG} and T_{op}	$R_{\theta JC}$
V dc	V (pk)	mW	°C	°C/W
100	150	250	-65 to +150	500

1/ Derate at 2.0 mW/°C above +25°C.

1.4 Primary electrical characteristics.

Limits	$V_{(BR)}$ I_R = 10 µA dc	R_S I_F = 100 mA dc 100 MHz	C_{VR} V_R = 100 V dc 100 kHz ≤ f ≤ 1 MHz	τ_{CL} I_F = 50 mA dc I_R = 250 mA dc
	V dc	ohms	pF	ns
Minimum	150			100
Maximum		1.25	.30	

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: NASA/Parts Project Office (NPPO), NASA Goddard Space Flight Center, Code 310.A, Greenbelt, MD 20771 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWING

Defense Electronics Supply Center

C68001 - Test Fixture for Effective Minority Carrier Lifetime.

(Copies of DESC drawing C68001 are available from Defense Electronics Supply Center, DESC-ECT, Dayton, OH 45444-5280.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

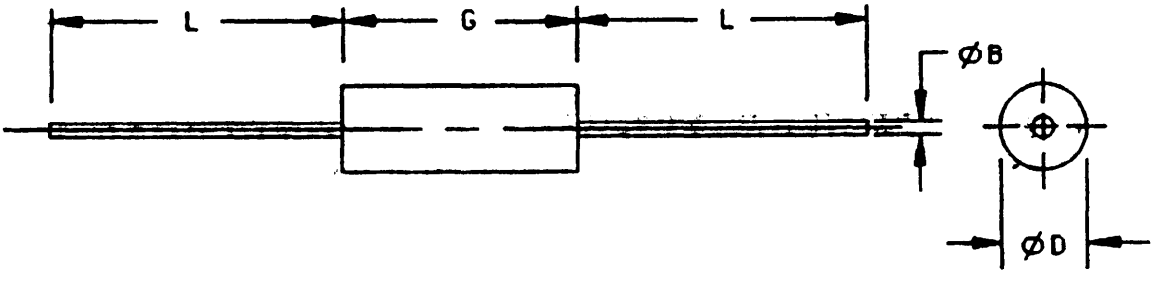
3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500 and as follows:

C_{VR}	Total capacitance with reverse bias.
τ_{CL}	Effective carrier lifetime.
P_{FM}	Peak forward power dissipation.
R_s	Residual series resistance.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500 and on figure 1 herein.

3.3.1 Lead finish. Lead finish shall be solderable as defined in MIL-S-19500, MIL-STD-750 and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

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Dimensions					
Ltr	Inches		Millimeters		Notes
	Min	Max	Min	Max	
G	.100	.170	2.54	4.32	3
ϕD	.045	.076	1.14	1.93	3
L	.975	1.500	24.77	38.10	
ϕB	.014	.016	0.36	0.41	4

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Package contour optional within ϕD and length G. Heat slugs, if any, shall be included within this cylinder but shall not be subject to minimum limit of ϕD.
4. Within this zone, lead diameter may vary to allow for lead finishes and irregularities other than heat slugs.

FIGURE 1. Physical dimensions.

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3.3.2 Polarity. The polarity shall be indicated by a contrasting color band to denote the cathode end. No color coding will be permitted.

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement
	JANTX and JANTXV
9	Not applicable
11	I_{R1} and V_{F1}
12	See 4.3.1
13	Subgroup 2 of table I herein, ΔI_{R1} = ±100 percent of initial value or 250 nA whichever is greater; ΔV_{F1} = 100 mV (peak).

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

T_A = room ambient as defined in the general requirements of MIL-STD-750, minimum; I_O = 70 mA dc, V_R = 120 V(pk), f = 60 Hz.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb (JANTX and JANTXV) of MIL-S-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.2.1 Group B inspection, table IVb (JANTX and JANTXV) of MIL-S-19500.

Subgroup	Method	Conditions
2	1056	Test condition A.
3	1027	I_O = 70 mA dc; V_R = 120 V (pk), f = 60 Hz. T_A = room ambient as defined in the general requirements of (see 4.5) of MIL-STD-750 (minimum).

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.3.1 Group C inspection, table V of MIL-S-19500.

Subgroup	Method	Conditions
2	2036	Test condition E.
6	1027	I_O = 70 mA dc; V_R = 120 V (pk), f = 60 Hz. T_A = room ambient as defined in the general requirements of MIL-STD-750 (minimum).

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions of pulse measurement shall be specified in section 4 of MIL-STD-750.

4.5.2 Effective carrier lifetime (τ_{CL}). Carrier lifetime may be measured using any of the following methods.

4.5.2.1 Reverse recovery time method. This test shall be conducted in accordance with MIL-STD-750, method 4031, condition B with the device installed in the test fixture, drawing C68001 or equivalent. The test parameters shall be: I_F = 50 mA, I_r = 30 mA, $I_{(rec)}$ = 3 mA.

4.5.2.2 Stored charge meter method. A pulse from a 20 kHz ± 200 Hz pulse generator and forward bias from a power supply shall be applied to the diode under test. Reverse current shall be 250 mA dc and forward bias shall be 50 mA dc. Read voltage on digital voltmeter and divide by 100 ohms to obtain average reverse current. Divide average reverse current by the product of pulse repetition frequency and bias current to obtain carrier lifetime (or equivalent) (see figure 2).

4.5.2.3 Stored charge meter method. Stored charge meter method (using commercial meter). Determine stored charge using meter manufacturer's instructions and a forward bias of 50 mA dc. Calculate the carrier lifetime by dividing the stored charge (coulombs) by the forward bias (amperes).

4.5.3 Series resistance (R_s). This test shall be performed utilizing an appropriate impedance analyzer such as the Hewlett Packard No. 4191A or a vector network analyzer such as the Hewlett Packard No. 8753B (or equivalent).

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Forward voltage	4011	I_F = 100 mA dc	V_{F1}		1.0	V dc
Reverse current	4016	DC method; V_R = 100 V dc	I_{R1}		250	nA dc
Breakdown voltage	4021	I_R = 10 µA dc	$V_{(BR)1}$	150		V dc
Subgroup 3						
High temperature operation:		T_A = +150°C				
Reverse current	4016	DC method; V_R = 100 V dc	I_{R2}		15	µA dc
Low temperature operation:		T_A = -65°C				
Forward voltage	4011	I_F = 100 mA dc	V_{F2}		1.15	V dc
Breakdown voltage	4021	I_R = 10 µA dc	$V_{(BR)2}$	150		V dc
Subgroup 4						
Capacitance	4001	V_R = 100 V dc; 100 kHz ≤ f ≤ 1 MHz	C		0.3	pF
Subgroups 5 and 6						
Not applicable						
Subgroup 7						
Series resistance		I_F = 100 mA dc 100 MHz ≤ f ≤ 1,000 MHz	R_s		1.25	ohms
Effective carrier lifetime		I_F = 50 mA dc (see 4.5.2)	τ_{CL}	100		ns

1/ For sampling plan, see MIL-S-19500.

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TABLE II. Groups B and C electrical measurements. 1/ 2/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Forward voltage	4011	I_F = 100 mA dc	V_{F1}		1.00	V dc
2.	Reverse current	4016	DC method; V_R = 100 V dc	I_{R1}		250	nA dc
3.	Breakdown voltage	4021	I_R = 10 µA dc	$V_{(BR)1}$	150		V dc
4.	Forward voltage	4011	I_F = 100 mA dc	ΔV_{F1}	±100 mV dc maximum		
5.	Reverse current	4016	DC method; V_R = 100 V dc	ΔI_{R1}	±25 percent of initial value or ± 50 nA dc (whichever is greater) change from initial group A reading.		

1/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1 and 2.

b. Subgroup 3, see table II herein, steps 1 and 5.

c. Subgroup 6, see table II herein, steps 1, 3, 4, and 5.

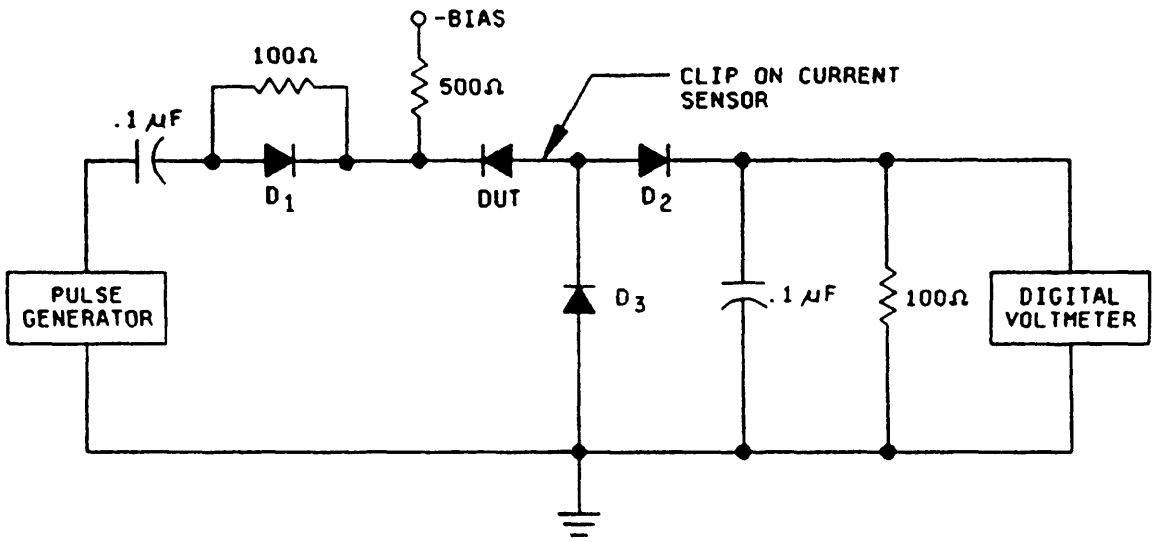
2/ The electrical measurements for table V of MIL-S-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1 and 2.

b. Subgroup 3, see table II herein, steps 1 and 2.

c. Subgroup 6, see table II herein, steps 1 and 2.

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NOTE: D1 and D2 shall be 1N4456 high conductive diodes, or equivalent. For D3, use Hewlett Packard No. 5082-2835, or equivalent.

FIGURE 2. Stored charge meter method.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish as specified (see 3.3.1).

c. Product assurance level and type designator.

6.3 Substitution information.

6.3.1 Substitution of JAN devices. JANTX devices are a one way direct substitute for JAN devices example; (JANTX1N5719 for JAN1N5719).

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Air Force - 19, 80, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC, OS, SH
Air Force - 13, 15, 70

Preparing activity:
NASA - NA

Agent:
DLA - ES

(Project 5961-1394)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/443A	2. DOCUMENT DATE (YYMMDD) 931020

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, DIODE, SILICON, SWITCHING TYPE 1N5719 JANTX AND JANTXV

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, first, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (if applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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